

Experiment 19: Student-Teacher-Sub-Code

Aim

To implement a simple **Student–Teacher knowledge base using Prolog** and query relationships such as who teaches a student, what subject a student studies, and who teaches a particular subject.

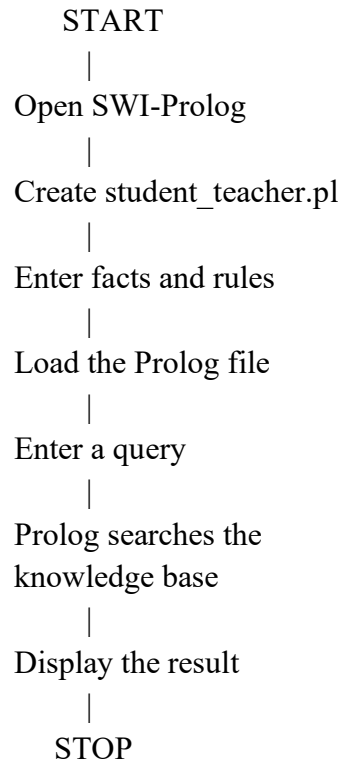
Objective

- To understand facts and rules in Prolog.
- To create a simple knowledge base for students, teachers, and subjects.
- To execute Prolog queries.
- To retrieve relationships using logical inference.

Algorithm

1. Start SWI-Prolog.
2. Create a Prolog file named student_teacher.pl.
3. Define facts for teachers and the subjects they teach.
4. Define facts for students and the subjects they study.
5. Define rules to determine:
 - a. The subject taught by a teacher.
 - b. The subject studied by a student.
 - c. The teacher who teaches a particular subject.
6. Load the Prolog file using consult.
7. Enter queries at the Prolog prompt.
8. Observe the results.
9. Stop.

Flowchart



Prolog Code

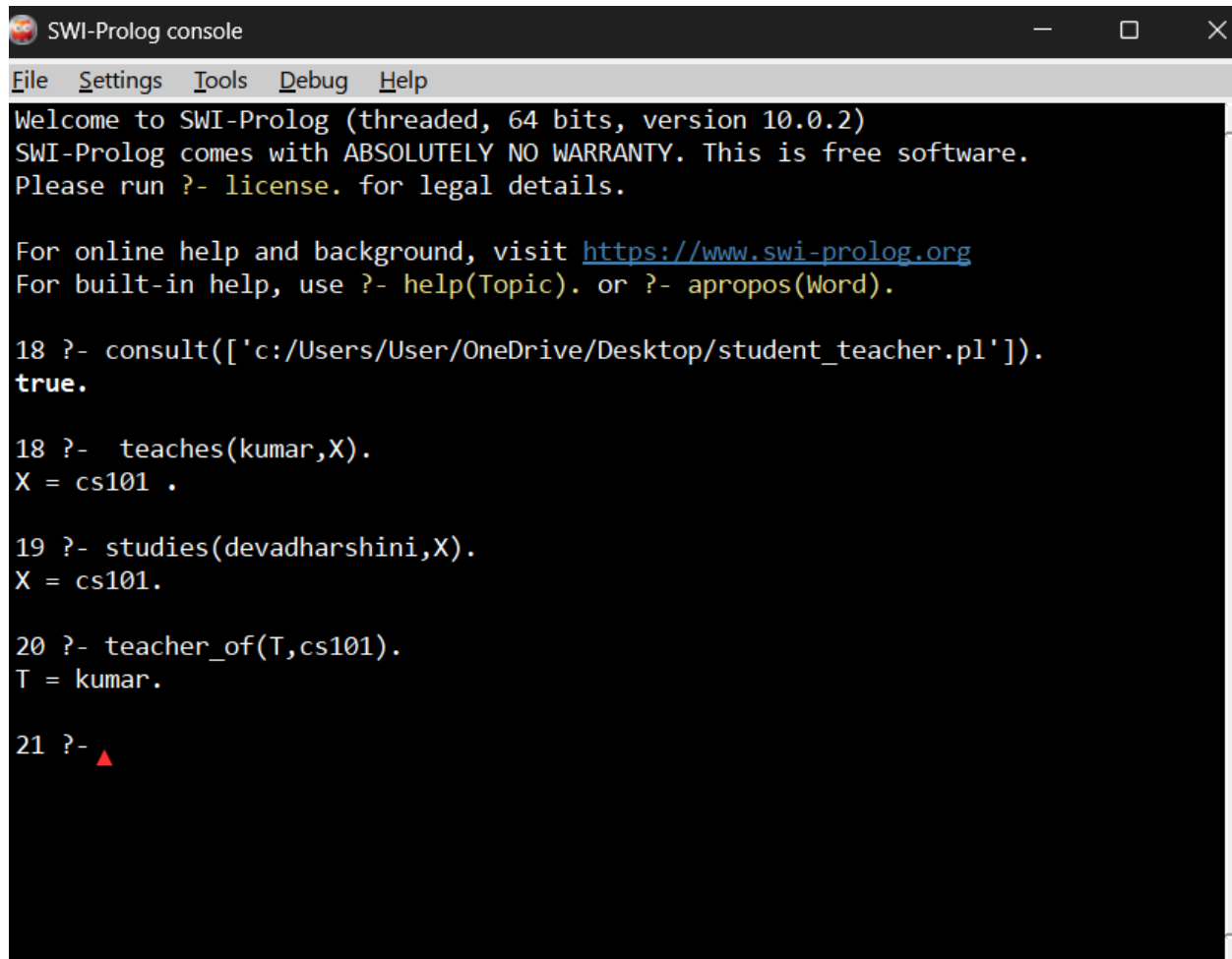
```
% Teacher and subject facts
teaches(kumar, cs101).
teaches(ravi, maths).
teaches(priya, physics).
```

```
% Student and subject facts
studies(devadharshini, cs101).
studies(anu, maths).
studies(rahul, physics).
```

```
% Rules
teacher_of(T, S) :-
    teaches(T, S).
```

```
student_of(S, Sub) :-
    studies(S, Sub).
```

Sample Output



```
SWI-Prolog console
File Settings Tools Debug Help
Welcome to SWI-Prolog (threaded, 64 bits, version 10.0.2)
SWI-Prolog comes with ABSOLUTELY NO WARRANTY. This is free software.
Please run ?- license. for legal details.

For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).

18 ?- consult(['c:/Users/User/OneDrive/Desktop/student_teacher.pl']).
true.

18 ?- teaches(kumar,X).
X = cs101 .

19 ?- studies(devadharshini,X).
X = cs101.

20 ?- teacher_of(T,cs101).
T = kumar.

21 ?- ▲
```

Result

The Student–Teacher knowledge base was successfully implemented in **SWI-Prolog**. The required relationships were obtained correctly using Prolog facts, rules, and queries.

Conclusion

Thus, the experiment successfully demonstrates the use of **Prolog for representing knowledge and performing logical queries**. The system can identify the subjects taught by teachers, subjects studied by students, and teachers associated with a particular subject.